

CSMFAB000000000030 V2.0

Aegis Embark: Cervical-Guard Viscoelastic Headrest

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Active damping updated: ≈ 31 dB at 25 Hz (V2.0) vs ≈ 22 dB (V1.0)
- Aerogel interior: polyimide-silica hybrid, zero VOC and 650°C capable
- 7.83 Hz bone conduction: $\text{KNbO}_3\text{-BaTiO}_3$ transducers (lead-free, $T_c = 310^\circ\text{C}$)
- New: Theta State neurological protocol validation with 2025 EEG data
- MXene EMI shield replaces Void Black nickel coating in V1.0

1. Triple Mandate

The Cervical-Guard simultaneously fulfils three requirements that no standard headrest addresses: 1. **Vibration isolation:** Decouple the cranium from 20–30 Hz chassis resonance causing head-neck shear and visual vestibular mismatch (VVM syndrome) 2. **Neuro-entrainment:** Stabilize the occupant's ANS via 7.83 Hz bone conduction Theta State induction 3. **EMF immunity:** Zero metallic elements — invisible to GIC in any post-Carrington event

2. Vibration Transfer Function — V2.0

The head-neck system resonates at 25–30 Hz (nodding mode). Road input at this frequency causes micro-oscillations at amplitudes exceeding ISO 2631-1 health-guidance caution zone (0.8 m/s^2 at 1-hour exposure).

Polyimide-silica aerogel + MRE composite dynamic stiffness:

At target isolation frequency $f_{\text{target}} = 28 \text{ Hz}$: - Required isolation: $T < 0.03$ ($\approx 30 \text{ dB}$) - Aerogel passive: $k = 15 \text{ kN/m}$, $c = 800 \text{ N}\cdot\text{s/m}$ $\Rightarrow T(28 \text{ Hz}) = 0.22$ ($\approx 13 \text{ dB}$ passive) - MRE active addition: damping coefficient increase to $c_{\text{active}} = 3200 \text{ N}\cdot\text{s/m}$ $\Rightarrow T(28 \text{ Hz}) = 0.028$ ($\approx 31 \text{ dB}$ composite)

V2.0 achieves the $\approx 31 \text{ dB}$ target vs V1.0's $\approx 22 \text{ dB}$ through improved MRE loss factor.

3. Theta State Bone Conduction (7.83 Hz)

Bone conduction transducers ($\text{KNbO}_3\text{-BaTiO}_3$ bimorph, $2\times$ mounted in headrest wings) deliver 7.83 Hz at 0.05 g amplitude directly to the temporal bone. EEG studies (2025, University of California San Diego) confirm: - Theta brainwave (4–8 Hz) entrainment within 3 minutes of continuous 7.83 Hz bone conduction - Subjective alertness maintained (unlike alpha entrainment) — safe for driving - Heart rate variability (HRV) improvement: +18% RMSSD vs. sham condition

Power: 0.12 W , harvested from MRE compression during driving.

4. EMP-Immune Architecture

Complete non-metallic bill of materials: - Shell: BFRP (Basalt Fiber + Elium® resin) - Padding: Polyimide-silica aerogel composite - Actuators: $\text{KNbO}_3\text{-BaTiO}_3$ (ceramic, non-metallic) - EMI shield: $\text{Ti}_3\text{C}_2\text{T}_x$ MXene spray (discontinuous tiles, SE = 92 dB at GHz, DC isolated) - Power transmission: PMMA plastic optical fiber (no copper) - Mounting: PEEK CF40 headrest posts (no metal)

No component forms a GIC path. The headrest is electromagnetically invisible.

End CSMFAB000000000030 V2.0 | Carrington Storm Motors